SUCCESSIVE DIFFERENTION

1. If $y = \frac{\log x}{x}$, show that $\frac{d^2y}{dx^2} = \frac{2\log x - 3}{x^3}$

2. If $y = \frac{a+bx}{c+dx}$, Show that $2y_1 y_3 = 3y_2^2$

3. If $y = \left(\frac{1}{x}\right)^x$, show that $y_2(1) = 0$.

4. If $x = a\left[\cos t + \log \tan \frac{t}{2}\right]$, $y = a\sin t$, find $\frac{d^2y}{dx^2}$

5. If $y = \left[x + \sqrt{x^2+1}\right]^n$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - n^2 y = 0$

6. If $y = x + \tan x$, show $\cos^2 x \frac{d^2y}{dx^2} - 2y + 2x = 0$.

DO ALL ABOVE QUESTIONS YOURSELF

\# Ex- If $ax^2 + 2hxy + by^2 = 1$, show $\frac{d^2y}{dx^2} = \frac{h^2 - ab}{(hx+by)^3}$

Sol:- Differentiating w.r.to x, we get

$$2ax + 2h(xy_1 + y) + 2byy_1 = 0$$

$$\Rightarrow y_1 = -\left(\frac{ax+hy}{hx+by}\right) \quad \text{—— (1)}$$

$$\Rightarrow y_2 = -\left[\frac{(hx+by)(a+hy_1) - (ax+hy)(h+by_1)}{(hx+by)^2}\right]$$

$$= \frac{(h^2-ab)(y-xy_1)}{(hx+by)^2} \quad \text{—— (2)}$$

$$y - xy_1 = y + x\left(\frac{ax+hy}{hx+by}\right) \quad \text{using (1)}$$

$$= \frac{1}{hx+by}$$

Substitute this in (2) we get required result.

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Ex- If $p^2 = a^2\cos^2\theta + b^2\sin^2\theta$, show $p + \frac{d^2p}{d\theta^2} = \frac{a^2b^2}{p^3}$

Sol:-

To show that $p^4 + p^3\frac{d^2p}{d\theta^2} = a^2b^2$

Diff. given eqn w.r.to. θ. we get-

$$2p\frac{dp}{d\theta} = 2a^2\cos\theta\cdot(-\sin\theta) + 2b^2\sin\theta\cos\theta$$

$$\Rightarrow p\frac{dp}{d\theta} = (b^2-a^2)\sin\theta\cos\theta \quad \text{---(1)}$$

Again diff. w.r.to θ.

$$\Rightarrow p\frac{d^2p}{d\theta^2} + \left(\frac{dp}{d\theta}\right)^2 = (b^2-a^2)(\cos^2\theta - \sin^2\theta)$$

$$\Rightarrow p\frac{d^2p}{d\theta^2} + \frac{(b^2-a^2)^2\sin^2\theta\cos^2\theta}{p^2} = (b^2-a^2)(\cos^2\theta - \sin^2\theta)$$

using (1)

$$\Rightarrow p^3\frac{d^2p}{d\theta^2} = (b^2-a^2)(\cos^2\theta - \sin^2\theta)(a^2\cos^2\theta + b^2\sin^2\theta) - (b^2-a^2)^2\sin^2\theta\cos^2\theta$$

$$= a^2b^2(\cos^4\theta + \sin^4\theta) - a^4\cos^4\theta - b^4\sin^4\theta$$

Now

$$\Rightarrow p^4 + p^3\frac{d^2p}{d\theta^2} = (a^2\cos^2\theta + b^2\sin^2\theta) + a^2b^2(\cos^4\theta + \sin^4\theta) - a^4\cos^4\theta - b^4\sin^4\theta$$

$$= 2a^2b^2\cos^2\theta\sin^2\theta + a^2b^2(\cos^4\theta + \sin^4\theta)$$

$$= a^2b^2[\cos^4\theta + \sin^4\theta + 2\cos^2\theta\sin^2\theta]$$

$$= a^2b^2(\cos^2\theta + \sin^2\theta)^2$$

$$\Rightarrow p^4 + p^3\frac{d^2p}{d\theta^2} = a^2b^2.$$

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STANDARD FORMS (N^{th} DERIVATIVE)

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S.F. I. $\boxed{y = \frac{1}{ax+b} = (ax+b)^{-1}}$

SOL:

$$y_1 = -1.(ax+b)^{-2}.a = \frac{-a}{(ax+b)^2}$$

$$y_2 = 2a(ax+b)^{-3}.a = \frac{2a^2}{(ax+b)^3}$$

$$y_3 = -2.3a^2(ax+b)^{-4}.a = \frac{-2.3a^3}{(ax+b)^4}.$$

$$\boxed{y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}}$$

S.F. II (a) - $\boxed{y = \sin(ax+b)}$

SOL:

$$y_1 = \cos(ax+b).a = a\sin\left[\frac{\pi}{2}+ax+b\right] \quad \left[\text{using } \cos\theta = \sin\left(\frac{\pi}{2}+\theta\right)\right]$$

$$y_2 = a\cos\left[\frac{\pi}{2}+ax+b\right].a = a^2\sin\left[\frac{\pi}{2}+\frac{\pi}{2}+ax+b\right] \quad ''$$

$$y_3 = a^2\cos\left[\frac{\pi}{2}+\frac{\pi}{2}+ax+b\right].a = a^3\sin\left[\frac{\pi}{2}+\frac{\pi}{2}+\frac{\pi}{2}+ax+b\right] \quad ''$$

$$\boxed{y_n = a^n \sin\left[\frac{n\pi}{2}+(ax+b)\right]}$$

II (b) - $\boxed{y = \cos(ax+b)}$ DO YOURSELF $\left[\text{using: } -\sin\theta = \cos\left(\frac{\pi}{2}+\theta\right)\right]$

SOL:- $\boxed{y_n = a^n\cos\left[\frac{n\pi}{2}+ax+b\right]}$

S.F. III (a) - $\boxed{y = e^{ax}\sin(bx+c)}$

SOL:

$$y_1 = e^{ax}.b\cos(bx+c) + ae^{ax}.\sin(bx+c)$$

Put $a = r\cos\theta$ & $b = r\sin\theta$ $\therefore r = \sqrt{a^2+b^2}$ & $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

$$\therefore y_1 = e^{ax}[r\sin\theta\cos(bx+c) + r\cos\theta\sin(bx+c)]$$

$$= re^{ax}\sin(bx+c+\theta)$$

$$y_2 = r\frac{d}{dx}\left[e^{ax}\sin(bx+c+\theta)\right]$$

$$= r.r.e^{ax}\sin(bx+c+\theta+\theta)$$

Contd----

$$y_2 = r^2 e^{ax} \sin(bx + c + 2\theta)$$

$$- - - - - - - - - -$$

$$y_n = r^n e^{ax} \sin(bx + c + n\theta)$$

$$\therefore \quad \boxed{y_n = (a^2+b^2)^{n/2} \cdot e^{ax} \sin\left[bx + c + n \tan^{-1}\left(\frac{b}{a}\right)\right]}$$

S.F III (b) $\quad \boxed{y = e^{ax}\cos(bx+c)}$

[DO YOURSELF]

$$\boxed{y_n = (a^2+b^2)^{n/2} \cdot e^{ax} \cos\left[bx + c + n \tan^{-1}\frac{b}{a}\right]}$$

1. Find the n^{th} derivatives of the following :

(i) $\frac{1}{x^2-4}$, (ii) $\frac{x}{x^2-3x+2}$, (iii) $\log(1-x^2)$.

Sol. (i) $y = \frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)} = \frac{1}{4}\left[\frac{1}{x-2} - \frac{1}{x+2}\right]$ $\left[\begin{array}{c}\text{using} \\ \frac{1}{(x+a)(x-a)} = \frac{1}{2a}\left[\frac{1}{x-a} - \frac{1}{x+a}\right]\end{array}\right]$

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$$\therefore \quad y_n = \frac{1}{4}\left[\frac{(-1)^n n!}{(x-2)^{n+1}} - \frac{(-1)^n n!}{(x+2)^{n+1}}\right]$$

$$= \frac{(-1)^n n!}{4}\left[\frac{1}{(x-2)^{n+1}} - \frac{1}{(x+2)^{n+1}}\right]$$

(ii) $y = \frac{x}{x^2-3x+2} = \frac{x}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$

$$\Rightarrow A(x-2) + B(x-1) = x \Rightarrow A = -1, \quad B = 2$$

$$\therefore \quad y = \frac{-1}{x-1} + \frac{2}{x-2} \Rightarrow y_n = \frac{d^n}{dx^n}\left(\frac{-1}{x-1}\right) + \frac{d^n}{dx^n}\left(\frac{2}{x-2}\right).$$

$$y_n = \frac{(-1)\cdot(-1)^n n!}{(x-1)^{n+1}} + \frac{2(-1)^n n!}{(x-2)^{n+1}}$$

(iii) $y = \log(1-x^2) \Rightarrow y_1 = \frac{-2x}{(1-x^2)} = \frac{A}{(1+x)} + \frac{B}{1-x}$

on solving we get $A = 1$, $B = -1$

$$\therefore \quad y_n = \frac{d^{n-1}}{dx^{n-1}}\left(\frac{1}{1+x}\right) + \frac{d^{n-1}}{dx^{n-1}}\left(\frac{1}{x-1}\right)$$

$$= \frac{(-1)^{n-1}(n-1)!}{(1+x)^n} + \frac{(-1)^{n-1}(n-1)!}{(x-1)^n} = (-1)^{n-1}(n-1)!\left[\frac{1}{(1+x)^n} + \frac{1}{(x-1)^n}\right]$$

2. Find the n^{th} derivatives of the following:

(i) $\sin^2 x$

let $y = \sin^2 x = \frac{1-\cos 2x}{2}$

$= \frac{1}{2} - \frac{1}{2}\cos 2x$

$y_n = 0 - \frac{1}{2}\frac{d^n}{dx^n}(\cos 2x)$

$= -\frac{1}{2}\cdot 2^n \cos\left(\frac{n\pi}{2} + 2x\right)$

(ii) $\sin^3 x$

let $y = \sin^3 x = \frac{3\sin x - \sin 3x}{4}$

$y_n = \frac{3}{4}\sin\left(\frac{n\pi}{2}+x\right) - \frac{1}{4}\cdot 3^n \sin\left(\frac{n\pi}{2}+3x\right.$

(iii) $\sin^4 x$

let $y = (\sin^2 x)^2 = \left(\frac{1-\cos 2x}{2}\right)^2$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{\cos^2 2x}{4}$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}\left(\frac{1+\cos 4x}{2}\right)$

do yourself.

(iv) $\sin 2x \cos x$

$y = \frac{1}{2}[\sin 3x + \sin x]$

do yourself

(v) $\cos^2 x$, (vi) $\cos^3 x$, (vii) $\cos^4 x$.

Do yourself.

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(viii) $y = e^{2x}\cos^2 x$

$= e^{2x}\left(\frac{1+\cos 2x}{2}\right)$

$= \frac{1}{2}e^{2x} + \frac{1}{2}e^{2x}\cos 2x$

$y_n = \frac{1}{2}\cdot 2^n e^{2x} + \frac{1}{2}e^{2x}\cdot(2^2+2^2)^{n/2}\cdot\cos\left[2x + n\tan^{-1}\left(\frac{2}{2}\right)\right]$

$= \cancel{\frac{1}{2}}\, 2^{n-1}\cdot e^{2x} + \frac{1}{2}e^{2x}\, 8^{n/2}\cos(2x + n\tan^{-1}1)$

$= 2^{n-1}e^{2x} + \frac{1}{2}\cdot 8^{n/2} e^{2x}\cos\left(2x + \frac{n\pi}{4}\right).$

(IX) $y = e^x \sin x \sin 2x$

$= \frac{e^x}{2}[\cos x - \cos 3x]$

$y_n = \frac{1}{2}e^x(1^2+1^2)^{n/2}\cos\left(x + n\tan^{-1}\left(\frac{1}{1}\right)\right) - \frac{1}{2}e^x(1^2+3^2)^{n/2}\cos\left[3x + n\tan^{-1}\frac{3}{1}\right.$

$= \frac{1}{2}e^x 2^{n/2}\cos\left(x + \frac{n\pi}{4}\right) - \frac{1}{2}e^x(10)^{n/2}\cos(3x + n\tan^{-1}3).$

IMPORTANT QUESTIONS

Q1:- Prove that the n^{th} derivative of $\frac{x^3}{x^2-1}$ at $x=0$ is ZERO if n is even and $-(n)!$ if n is odd $(n>1)$.

Sol:- let $y = \frac{x^3}{x^2-1} = x + \frac{x}{x^2-1} = x + \frac{1}{2}\left(\frac{1}{x-1} + \frac{1}{x+1}\right)$

$$y_n = 0 + \frac{1}{2}\left[\frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{(-1)^n n!}{(x+1)^{n+1}}\right]$$

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$$= \frac{1}{2}(-1)^n n! \left[\frac{1}{(x-1)^{n+1}} + \frac{1}{(x+1)^{n+1}}\right]$$

$$y_n(0) = \frac{1}{2}(-1)^n n! \left[\frac{1}{(-1)^{n+1}} + 1\right]$$

Case ① n is even : $y_n(0) = \frac{1}{2}(-1)^n n! \,(-1+1) = 0$

Case ② n is odd : $y_n(0) = \frac{1}{2}(-1)^n n!\,(1+1) = (-1)^n n! = -(n)!$

NOTE: $$\frac{d^n}{dx^n}\left[\log(ax+b)\right] = \frac{(-1)^{n-1}(n-1)!\,a^n}{(ax+b)^n}$$

Q2:- If $y = x\log\left(\frac{x-1}{x+1}\right)$, prove that

$$\frac{d^n y}{dx^n} = (-1)^n (n-2)!\left[\frac{(x-n)}{(x-1)^n} - \frac{x+n}{(x+1)^n}\right]$$

Sol:- $y = x\log(x-1) - x\log(x+1)$

$$y_1 = \frac{x}{x-1} + \log(x-1) - \frac{x}{x+1} - \log(x+1)$$

$$= \frac{1}{x-1} + \frac{1}{x+1} + \log(x-1) - \log(x+1)$$

Applying $(n-1)^{th}$ derivative, we get

$$y_n = (-1)^{n-1}(n-1)!\left[\frac{1}{(x-1)^n} + \frac{1}{(x+1)^n}\right] + (-1)^{n-2}(n-2)!\left[\frac{1}{(x-1)^{n-1}} - \frac{1}{(x+1)^{n-1}}\right]$$

$$= (-1)^n (n-2)!\left[\frac{-(n-1)}{(x-1)^n} - \frac{(n-1)}{(x+1)^n} + \frac{1}{(x-1)^{n-1}} - \frac{1}{(x+1)^{n-1}}\right]$$

$$= (-1)^n (n-2)!\left[\frac{-n+1+x-1}{(x-1)^n} - \frac{n-1+x+1}{(x+1)^n}\right]$$

After solving we get the reqd. proof.

Q3:- If $y = \sin nx + \cos nx$, show that:

$$y_r = n^r\sqrt{1+(-1)^r \sin 2nx}$$

Sol:-
$$y_r = n^r \sin\left(nx + \frac{r\pi}{2}\right) + n^r \cos\left(nx + \frac{r\pi}{2}\right)$$
$$= n^r \left[\sin\left(nx + \frac{r\pi}{2}\right) + \cos\left(nx + \frac{r\pi}{2}\right)\right]$$
$$= n^r \sqrt{\left[\sin\left(nx + \frac{r\pi}{2}\right) + \cos\left(nx + \frac{r\pi}{2}\right)\right]^2}$$
$$= n^r \sqrt{1 + 2\sin\left(nx + \frac{r\pi}{2}\right)\cos\left(nx + \frac{r\pi}{2}\right)}$$
$$= n^r \sqrt{1 + \sin\left(2nx + r\pi\right)} = n^r\sqrt{1+(-1)^r \sin 2nx}$$

Q4:- If $I_n = \frac{d^n}{dx^n}(x^n \log x)$, prove $I_n = nI_{n-1} + (n-1)!$

Deduce that $I_n = n!\left(\log x + 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right)$

Sol:-
$$I_n = \frac{d^n}{dx^n}(\log x \cdot x^n) = \frac{d^{n-1}}{dx^{n-1}}\left[\frac{d}{dx}(x^n \log x)\right]$$
$$= \frac{d^{n-1}}{dx^{n-1}}\left[\frac{x^n}{x} + nx^{n-1}\log x\right] = \frac{d^{n-1}}{dx^{n-1}}(x^{n-1}) + n\frac{d^{n-1}}{dx^{n-1}}(x^{n-1}\log x)$$
$$= (n-1)! + nI_{n-1} \quad \text{--- (1)}$$

Now (1) $\Rightarrow$
$$I_{n-1} = (n-2)! + (n-1)I_{n-2}$$
$$I_{n-2} = (n-3)! + (n-2)I_{n-3}$$
$$\text{------}$$
$$I_3 = 2! + 3I_2$$
$$I_2 = 1! + 2I_1$$

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and $I_1 = \frac{d}{dx}(x\log x) = 1 + \log x$

Substitute these values successively in (1) and simplifying, we get required result.

NOTE :- $\dfrac{d^n}{dx^n}\left(\dfrac{1}{ax+b}\right) = \dfrac{(-1)^n\, n!\, a^n}{(ax+b)^{n+1}}$

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:- $\dfrac{1}{(x+a)(x-a)} = \dfrac{1}{2a}\left(\dfrac{1}{x-a} - \dfrac{1}{x+a}\right)$

:- $\dfrac{x}{(x+a)(x-a)} = \dfrac{1}{2}\left(\dfrac{1}{x-a} + \dfrac{1}{x+a}\right)$

Special n^{th} derivatives (Based on Demoivre's Thm)

1:- Find n^{th} derivative of $\dfrac{1}{x^2+a^2}$

Sol:- $y = \dfrac{1}{x^2+a^2} = \dfrac{1}{(x+ai)(x-ai)} = \dfrac{1}{2ai}\left(\dfrac{1}{x-ai} - \dfrac{1}{x+ai}\right)$

$$y_n = \frac{1}{2ai}(-1)^n n!\left(\frac{1}{(x-ai)^{n+1}} - \frac{1}{(x+ai)^{n+1}}\right) \quad \text{——}(1)$$

Now $x+ai = r(\cos\theta + i\sin\theta)$, $r = \sqrt{x^2+a^2}$, $\theta = \tan^{-1}\left(\frac{a}{x}\right)$

$\therefore\ (x+ai)^{n+1} = r^{n+1}\left[\cos(n+1)\theta + i\sin(n+1)\theta\right]$

||ly $(x-ai)^{n+1} = r^{n+1}\left(\cos(n+1)\theta - i\sin(n+1)\theta\right)$

$(1) \Rightarrow\ y_n = \dfrac{(-1)^n n!}{2ai}\left[\dfrac{1}{r^{n+1}}\{\cos(n+1)\theta + i\sin(n+1)\theta\} - \dfrac{1}{r^{n+1}}\{\cos(n+1)\theta - i\sin(n+1)\theta\}\right]$

$$= \frac{(-1)^n n!}{\cancel{2}a\cancel{i}\cdot r^{n+1}}\,\cancel{2}\cancel{i}\sin(n+1)\theta = \frac{(-1)^n n! \,.\, \sin(n+1)\theta}{a\,(x^2+a^2)^{\frac{n+1}{2}}}$$

$\theta = \tan^{-1}\left(\frac{a}{x}\right)$

---

2. Find the n^{th} derivative of $\dfrac{1}{x^2+x+1}$

Sol. let $y = \dfrac{1}{x^2+x+1} = \dfrac{1}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

Now by Q1. we have $x = \left(x+\frac{1}{2}\right)$, $a = \frac{\sqrt{3}}{2}$

we get

$$y_n = \frac{(-1)^n n!\ \sin(n+1)\theta}{\frac{\sqrt{3}}{2}\left(x^2+x+1\right)^{\frac{n+1}{2}}}$$

where $\theta = \tan^{-1}\dfrac{\sqrt{3}/2}{x+\frac{1}{2}}$

$= \tan^{-1}\left(\dfrac{\sqrt{3}}{2x+1}\right).$

3. Find the n^{th} derivative of $\frac{x}{x^2+a^2}$

Sol. let $y = \frac{x}{x^2+a^2} = \frac{x}{(x+ai)(x-ai)} = \frac{1}{2}\left[\frac{1}{x-ai} + \frac{1}{x+ai}\right]$

Now proceed as Q1. we get

$$y_n = \frac{(-1)^n n!}{(x^2+a^2)^{\frac{n+1}{2}}} \cos(n+1)\theta, \text{ where } \theta = \tan^{-1}\left(\frac{a}{x}\right).$$

---

4. If $y = \tan^{-1}x$, show : $y_n = (-1)^{n-1}(n-1)!\,\sin n\left(\frac{\pi}{2}-y\right)\cdot\sin^n\left(\frac{\pi}{2}-y\right)$.

Sol. $y = \tan^{-1}x \Rightarrow y_1 = \frac{1}{1+x^2} = \frac{1}{(x+i)(x-i)} = \frac{1}{2i}\left[\frac{1}{x-i} - \frac{1}{x+i}\right]$

Diff. w.r.to x $(n-1)$ times, we get,

$$y_n = \frac{1}{2i}(-1)^{n-1}(n-1)!\left[\frac{1}{(x-i)^n} - \frac{1}{(x+i)^n}\right] \quad \text{—(1)}$$

Now $x+i = r(\cos\theta + i\sin\theta)$ where $r = \sqrt{x^2+1}$, $\theta = \tan^{-1}\left(\frac{1}{x}\right)$.

$(x+i)^n = r^n(\cos n\theta + i\sin n\theta)$

$(x-i)^n = r^n(\cos n\theta - i\sin n\theta)$

$\therefore$ (1) $\Rightarrow y_n = \frac{1}{2i}(-1)^{n-1}(n-1)!\cdot\frac{1}{r^n}\left[\cos n\theta + i\sin n\theta - \cos n\theta + i\sin n\theta\right]$

$\Rightarrow y_n = \frac{(-1)^{n-1}(n-1)!}{(x^2+1)^{\frac{n}{2}}}.\sin n\theta$ where $\theta = \tan^{-1}\left(\frac{1}{x}\right)$. —(2)

Now

$y = \tan^{-1}x$ & $\theta = \tan^{-1}\left(\frac{1}{x}\right) \Rightarrow \frac{1}{x} = \tan\theta \Rightarrow x = \cot\theta$

So $y = \tan^{-1}(\cot\theta) = \frac{\pi}{2}-\theta \Rightarrow \theta = \frac{\pi}{2}-y$

Also $(x^2+1)^{n/2} = (\cot^2\theta+1)^{n/2} = (\operatorname{cosec}^2\theta)^{n/2} = \operatorname{cosec}^n\theta = \operatorname{cosec}^n\left(\frac{\pi}{2}-y\right)$

So After substituting these values we get reqd. result.

---

5. Find n^{th} derivative of (a) $\tan^{-1}\frac{2x}{1-x^2}$, (b) $\tan^{-1}\left(\frac{1+x}{1-x}\right)$, (c) $\log(x^2+a^2)$.

Sol. (a) $y = \tan^{-1}\frac{2x}{1-x^2}$, let $x = \tan\theta$

$y = 2\theta = 2\tan^{-1}x \Rightarrow y_1 = \frac{2}{1+x^2}$

$y_1 = \frac{2}{(x+i)(x-i)} = \frac{2}{2i}\left[\frac{1}{x-i} - \frac{1}{x+i}\right]$

do yourself

(b) $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$ let $x = \tan\theta$

$y = \frac{\pi}{4}+\theta = \frac{\pi}{4}+\tan^{-1}x \Rightarrow y_1 = \frac{1}{1+x^2}$

do your self.

(c) $y = \log(x^2+a^2)$

$y_1 = \frac{2x}{x^2+a^2} = \frac{2}{2}\left[\frac{1}{x-ai} + \frac{1}{x+ai}\right]$

do your self.

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LEIBNITZ THEOREM

STATEMENT : If U and V are two functions of x possessing n^{th} order derivatives, then:

$$(UV)_n = U_n V + {}^nC_1 U_{n-1} V_1 + {}^nC_2 U_{n-2} V_2 + \text{-------} + U V_n$$

1. If $y = x^2 \sin x$, prove that

$$y_n = (x^2 - n^2 + n)\sin\left(x + \frac{n\pi}{2}\right) - 2nx\cos\left(x + \frac{n\pi}{2}\right)$$

Sol. let $U = \sin x$ and $V = x^2$

$U_n = \sin\left(x + \frac{n\pi}{2}\right)$, $V_1 = 2x$

$U_{n-1} = \sin\left(x + \frac{(n-1)\pi}{2}\right)$, $V_2 = 2.$

By Leibnitz theorem

$$y_n = x^2 \sin\left(x + \frac{n\pi}{2}\right) + {}^nC_1 \sin\left(x + \frac{(n-1)\pi}{2}\right)\cdot 2x + {}^nC_2 \sin\left(x + \frac{(n-2)\pi}{2}\right)\cdot 2$$

$$= x^2 \sin\left(x + \frac{n\pi}{2}\right) - 2nx \sin\left[\frac{\pi}{2} - \left(x + \frac{n\pi}{2}\right)\right] \mp \frac{n(n-1)}{2}\cdot \sin\left[\pi - \left(x + \frac{n\pi}{2}\right)\right]$$

$$= x^2 \sin\left(x + \frac{n\pi}{2}\right) - 2nx\cos\left(x + \frac{n\pi}{2}\right) - (n^2 - n)\sin\left(x + \frac{n\pi}{2}\right)$$

$$\therefore\ y_n = (x^2 - n^2 + n)\sin\left(x + \frac{n\pi}{2}\right) - 2nx\cos\left(x + \frac{n\pi}{2}\right).$$

2. If $y = e^{m\sin^{-1}x}$, show :-

$$(1 - x^2)y_{n+2} - (2n+1)x\,y_{n+1} - (n^2 + m^2)y_n = 0$$

Sol. $y = e^{m\sin^{-1}x} \Rightarrow y_1 = \frac{e^{m\sin^{-1}x}}{\sqrt{1-x^2}}\cdot m = \frac{my}{\sqrt{1-x^2}}$

$\Rightarrow y_1\sqrt{1-x^2} = my \Rightarrow y_1^2(1-x^2) = m^2y^2$

$\Rightarrow 2y_1y_2(1-x^2) + y_1^2(-2x) = 2m^2yy_1$

$\Rightarrow y_2(1-x^2) - xy_1 = m^2y$ (Cancelling $2y_1$).

Diff. w.r.to x n times, we get

$$y_{n+2}(1-x^2) + {}^nC_1 y_{n+1}(-2x) + {}^nC_2 y_n(-2).$$

$$- \left[x\,y_{n+1} + {}^nC_1 y_n\cdot 1\right] = m^2 y_n$$

${}^nC_1 = n$, ${}^nC_2 = \frac{n(n-1)}{2\cdot 1}$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+m^2)y_n = 0.$$

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3. If $y = \cos(m \sin^{-1} x)$, show :
$$(1-x^2)y_{n+2} - (2n+1)x\,y_{n+1} + (m^2-n^2)y_n = 0$$

Sol. $y_1 = -\sin(m\sin^{-1}x) \cdot \dfrac{m}{\sqrt{1-x^2}} = \dfrac{-m\sqrt{1-\cos^2(m\sin^{-1}x)}}{\sqrt{1-x^2}}$

$\Rightarrow \sqrt{1-x^2}\, y_1 = m\sqrt{1-y^2} \Rightarrow (1-x^2)y_1^2 = m^2 - m^2y^2$

$\Rightarrow 2y_1y_2(1-x^2) + y_1^2(-2x) = -2m^2 y y_1$

$\Rightarrow y_2(1-x^2) - xy_1 + m^2 y = 0$ Do yourself.

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4. If $y = \left[x + \sqrt{1+x^2}\right]^m$, show :
$$(x^2+1)y_{n+2} + (2n+1)x\,y_{n+1} + (n^2-m^2)y_n = 0$$

Sol. $y_1 = m\left[x+\sqrt{1+x^2}\right]^{m-1}\left[1 + \dfrac{\cancel{2}x}{\cancel{2}\sqrt{1+x^2}}\right]$

$= \dfrac{m\left[x+\sqrt{1+x^2}\right]^m}{\left[x+\sqrt{1+x^2}\right]} \cdot \dfrac{\left[x+\sqrt{1+x^2}\right]}{\sqrt{1+x^2}} = \dfrac{my}{\sqrt{1+x^2}}$

$\Rightarrow \sqrt{1+x^2}\, y_1 = my \Rightarrow (1+x^2)y_1^2 = m^2y^2$

$\Rightarrow y_2(1+x^2) + y_1 x = m^2 y$ do yourself

5. If $y = (x^2-1)^n$, prove :
$$(x^2-1)y_{n+2} + 2x\,y_{n+1} - n(n+1)y_n = 0$$

Sol. $y_1 = \dfrac{2nxy}{x^2-1}$ (verify it) $\Rightarrow (x^2-1)y_1 = 2nxy$

$\Rightarrow y_2(x^2-1) + y_1 \cdot 2x = 2n(y + xy_1)$

$\Rightarrow y_2(x^2-1) + 2(1-n)y_1 x - 2ny = 0.$ do yourself

6. If $y = a\cos(\log x) + b\sin(\log x)$, show :
$$x^2 y_{n+2} + (2n+1)x\,y_{n+1} + (n^2+1)y_n = 0.$$

Sol. $y_1 x = -a\sin(\log x) + b\cos(\log x)$

$y_2 x + y_1 + y = 0$ Now do yourself.

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7. If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$, show:

$$x^2 y_{n+2} + (2n+1)x y_{n+1} + 2n^2 y_n = 0$$

Sol. $\cos^{-1}\left(\frac{y}{b}\right) = n\log\left(\frac{x}{n}\right)$

$\Rightarrow -\frac{1}{\sqrt{1-\frac{y^2}{b^2}}}\cdot\frac{1}{b}\cdot y_1 = \cancel{n}\frac{1}{x/n}\cdot\frac{1}{\cancel{n}} \Rightarrow -\frac{\cancel{b}}{\sqrt{b^2-y^2}}\cdot\frac{1}{\cancel{b}}\cdot y_1 = \frac{n}{x}$

$\Rightarrow -y_1\cdot x = n\sqrt{b^2-y^2} \Rightarrow y_1^2x^2 = n^2b^2 - n^2y^2$

$\Rightarrow 2y_1y_2\cdot x^2 + y_1^2\cdot 2x = -2n^2 y y_1$

$\Rightarrow y_2x^2 + y_1\cdot x + n^2 y = 0$ do yourself.

8. If $y = \frac{\sin^{-1}x}{\sqrt{1-x^2}}$. Show:

$$(1-x^2)y_{n+2} - (2n+3)x y_{n+1} - (n+1)^2 y_n = 0$$

Sol. $y\sqrt{1-x^2} = \sin^{-1}x \Rightarrow y^2(1-x^2) = (\sin^{-1}x)^2$

$\Rightarrow 2yy_1(1-x^2) + y^2(-2x) = \frac{2\sin^{-1}x}{\sqrt{1-x^2}} = 2y$

$\Rightarrow y_1(1-x^2) - xy = 1$ (Cancelling $2y$)

$\Rightarrow y_2(1-x^2) + y_1(-2x) - (y_1x + y) = 0$

$\Rightarrow y_2(1-x^2) - 3y_1x - y = 0$

Diff. 'n' times w.r.to. x, we get

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$\Rightarrow y_{n+2}(1-x^2) + {}^nC_1 y_{n+1}(-2x) + {}^nC_2 y_n(-2) - 3y_{n+1}\cdot x - 3\cdot{}^nC_1 y_n\cdot 1 - y_n = 0$

$\Rightarrow (1-x^2)y_{n+2} - (2n+3)x y_{n+1} - (n+1)^2 y_n = 0.$

9. If $y^{1/m} + y^{-1/m} = 2x$, prove:

$$(x^2-1)y_{n+2} + (2n+1)x y_{n+1} + (n^2-m^2)y_n = 0$$

Sol. let $u = y^{1/m}$ Then $u + \frac{1}{u} = 2x \Rightarrow u^2 - 2xu + 1 = 0$

$u = \frac{2x \pm\sqrt{4x^2-4}}{2} = x \pm \sqrt{x^2-1}$

Case ① $u = x + \sqrt{x^2-1}$ i.e. $y = \left[x + \sqrt{x^2-1}\right]^m$ proceed as in Q4

Case ② $u = x - \sqrt{x^2-1}$ i.e. $y = \left[x - \sqrt{x^2-1}\right]^m$

10. If $y = e^{m\cos^{-1}x}$, find $y_n(0)$

Sol. $y = e^{m\cos^{-1}x}$ proceeding as in ② we get,

$y_1 = \frac{-my}{\sqrt{1-x^2}}$ — ②

$y_2(1-x^2) - y_1 x = m^2 y$ — ③

$(1-x^2)y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+m^2)y_n = 0$ — ④

Now ① $\Rightarrow$ $y(0) = e^{\frac{m\pi}{2}}$

② $\Rightarrow$ $y_1(0) = -m\,y(0) = -me^{\frac{m\pi}{2}}$

③ $\Rightarrow$ $y_2(0) = m^2 y(0) = m^2 e^{\frac{m\pi}{2}}$

& ④ $\Rightarrow$ $y_{n+2}(0) = (n^2+m^2)\,y_n(0)$

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Putting $n = 1,3,5,---$, we get:

$y_3(0) = (1^2+m^2)\,y_1(0) = -(1^2+m^2)\,me^{\frac{m\pi}{2}}$

$y_5(0) = (3^2+m^2)\,y_3(0) = -me^{\frac{m\pi}{2}}(1^2+m^2)(3^2+m^2)$

— — — — — — — — — — — — — —

$y_{2k+1}(0) = -me^{\frac{m\pi}{2}}(1^2+m^2)(3^2+m^2)\cdots[(2k-1)^2+m^2]$ — Ⓐ

Putting $n = 2,4,6,\ldots$ we get

$y_4(0) = (2^2+m^2)\,y_2(0) = (2^2+m^2)\,m^2 e^{\frac{m\pi}{2}}$

$y_6(0) = (4^2+m^2)\,y_4(0) = m^2 e^{\frac{m\pi}{2}}(2^2+m^2)(4^2+m^2)$

— — — — — — — — — — — — — —

$y_{2k}(0) = m^2 e^{\frac{m\pi}{2}}(2^2+m^2)(4^2+m^2)\cdots[(2k-2)^2+m^2]$ — Ⓑ

Ⓐ and Ⓑ constitute the answer.

:... If $y = e^{m\sin^{-1}x}$, find $y_n(0)$ DO YOURSELF

11. If $y = \log\left[x+\sqrt{1+x^2}\right]$, find $y_n(0)$.

Sol. $y = \log[x+\sqrt{1+x^2}]$ — ①

$y_1 = \frac{1}{\sqrt{1+x^2}}$ — ② $y_2(1+x^2) + y_1 x = 0$ — ③

$(1+x^2)y_{n+2} + (2n+1)x\,y_{n+1} + n^2 y_n = 0$ — ④

① $\Rightarrow$ $y(0) = 0$, ② $\Rightarrow$ $y_1(0) = 1$, ③ $\Rightarrow$ $y_2(0) = 0$, ④ $\Rightarrow$ $y_{n+2}(0) = -n^2 y_n(0)$.

Putting $n = 2,4,6,--$ we get $y_4(0) = y_6(0) = ---- = 0$ ie $y_{2k}(0) = 0$ — Ⓐ

Contd.........

putting $n = \cancel{2,4,6}, \ldots$ $1, 3, 5, \ldots$ we get

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$$y_3(0) = -1^2 y_1(0) = -1^2$$
$$y_5(0) = -3^2 y_3(0) = 1^2 . 3^2$$
$$y_7(0) = -5^2 y_5(0) = -1^2 . 3^2 . 5^2$$
$$\text{- - - - - - - - - - - - - - -}$$
$$y_{2k+1}(0) = (-1)^k . 1^2 . 3^2 . 5^2 \ldots\ldots (2k-1)^2 \quad \text{——} \; (B)$$

(A) and (B) constitute the answer.

SOME IMPORTANT QUESTIONS

12. If $f(x) = \tan x$, prove that

$$f^n(0) - {}^nC_2 f^{n-2}(0) + {}^nC_4 f^{n-4}(0) + \cdots\cdots = \sin \frac{n\pi}{2}$$

Sol. $f(x) = \tan x = \dfrac{\sin x}{\cos x} \Rightarrow f(x) \cos x = \sin x$

$$\Rightarrow \frac{d^n}{dx^n}\left[f(x) \cos x\right] = \frac{d^n}{dx^n}(\sin x)$$

using Leibnitz thm

$$\Rightarrow f^n(x) . \cos x + {}^nC_1 f^{n-1}(x) . (-\sin x) + {}^nC_2 f^{n-2}(x)(-\cos x) + \cdots\cdots = \sin\left(x + \frac{n\pi}{2}\right)$$

Putting $x = 0$, we get

$$f^n(0) - {}^nC_2 f^{n-2}(0) + {}^nC_4 f^{n-4}(0) \cdots\cdots = \sin \frac{n\pi}{2}$$

---

13. Show that:

$$\frac{d^n}{dx^n}\left(\frac{\log x}{x}\right) = \frac{(-1)^n n!}{x^{n+1}}\left(\log x - 1 - \frac{1}{2} - \frac{1}{3} - \cdots\cdots - \frac{1}{n}\right)$$

Sol. let $U = \dfrac{1}{x}$, $V = \log x$

$$U_1 = -\frac{1}{x^2} \qquad V_1 = \frac{1}{x}$$
$$U_2 = \frac{2}{x^3} \qquad V_2 = -\frac{1}{x^2}$$
$$U_3 = -\frac{2.3}{x^4} \qquad V_3 = \frac{1.2}{x^3}$$
$$\text{- - - - - -} \qquad \text{- - - - - -}$$
$$U_n = \frac{(-1)^n n!}{x^{n+1}} \qquad V_n = \frac{(-1)^{n-1}(n-1)!}{x^n}$$

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By Leibnitz Thm $(UV)_n = U_n V + {}^nC_1 U_{n-1} V_1 + {}^nC_2 U_{n-2} V_2 + \cdots\cdots + U V_n$

LHS

$$= \frac{(-1)^n n!}{x^{n+1}} . \log x + n . \frac{(-1)^{n-1}(n-1)!}{x^n} . \frac{1}{x} + \frac{n(n-1)}{2!} \frac{(-1)^{n-2}(n-2)!}{x^{n-1}} . \left(-\frac{1}{x^2}\right) + \cdots + \frac{1}{x}\frac{(-1)^{n-1}(n-1)!}{x^n}$$

$$= \frac{(-1)^n . n!}{x^{n+1}}\left(\log x - 1 - \frac{1}{2} - \frac{1}{3} - \cdots\cdots - \frac{1}{n}\right) = \text{RHS}.$$

14. If $x+y=1$, prove that:

$$\frac{d^n}{dx^n}(x^n y^n) = n!\left[y^n - ({}^nc_1)^2 y^{n-1}x + ({}^nc_2)^2 y^{n-2}x^2 + \cdots + (-1)^n x^n\right]$$

Sol. let $U = x^n$ and $V = y^n = (1-x)^n$

$U_1 = nx^{n-1}$ $\quad V_1 = n(1-x)^{n-1}(-1)$

$U_2 = n(n-1)x^{n-2}$ $\quad V_2 = n(n-1)(1-x)^{n-2}$

$U_3 = n(n-1)(n-2)x^{n-3}$

$U_r = n(n-1)(n-2)\cdots(n-r+1)x^{n-r}$ $\quad V_r = \frac{(-1)^r n!}{(n-r)!}(1-x)^{n-r}$

$= \frac{n!}{(n-r)!}x^{n-r}$

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By Leibnitz thm

$$LHS = (UV)_n = U_nV + nU_{n-1}V_1 + \frac{n(n-1)}{2!}U_{n-2}V_2 + \cdots + UV_n$$

$$= \frac{n!}{0!}x^0(1-x)^n + n.\frac{n!}{1!}x^1.-n(1-x)^{n-1} + \frac{n(n-1)}{2!}\frac{n!}{2!}x^2.n(n-1)(1-x)^{n-2} + \cdots$$

$$\cdots + x^n.\frac{(-1)^n n!}{0!}(1-x)^0$$

$$= n!\left[y^n - {}^nc_1 y^{n-1}.x + {}^nc_2 y^{n-2}.x^2 - \cdots + (-1)^n x^n\right] = RHS.$$

15. If $y = e^{\frac{1}{2}x^2}.\cos x$, show that:

$$y_{2n+2}(0) - 4n\,y_{2n}(0) + 2n(2n-1)y_{2n-2}(0) = 0$$

Sol. $y_1 = e^{\frac{1}{2}x^2}(-\sin x) + e^{\frac{1}{2}x^2}.x\cos x$

$= -e^{\frac{1}{2}x^2}\sin x + y.x$ ——— (1)

$y_2 = -e^{\frac{1}{2}x^2}\cos x - e^{\frac{1}{2}x^2}.x\sin x + y.1 + y_1x$

$= \cancel{-y} + x(y_1 - yx) + \cancel{y} + y_1x$ $\quad$ using (1)

ie $y_2 - 2y_1x + yx^2 = 0$

Diff. 2n times (using Leibnitz thm), we get:

$$y_{2n+2} - 2\left[y_{2n+1}.x + {}^{2n}c_1 y_{2n}.1\right] + \left[y_{2n}.x^2 + {}^{2n}c_1 y_{2n-1}.2x + \frac{2n(2n-1)}{2!}y_{2n-2}.2\right] = 0$$

Putting $x=0$, we get

$$y_{2n+2}(0) - 4n\,y_{2n}(0) + 2n(2n-1)y_{2n-2}(0) = 0.$$

16. Find (i) $(x^2 \sin x)_n$ (ii) $(x^2 e^x \cos x)_n$

Sol. (i) $(\sin x)_n = \sin\left(x + \frac{n\pi}{2}\right)$

using Leibnitz Thm.

$$(x^2 \sin x)_n = (\sin x)_n\, x^2 + {}^nC_1 (\sin x)_{n-1} \cdot 2x + (\sin x)_{n-2} \cdot 2$$

$$= \sin\left(x + n\frac{\pi}{2}\right) \cdot x^2 + n \cdot \sin\left[x + \frac{(n-1)\pi}{2}\right] \cdot 2x + \frac{n(n-1)}{\cancel{2}} \sin\left[x + \frac{(n-2)\pi}{2}\right] \cdot \cancel{2}$$

(ii). $$\left[e^{ax}\cos(bx+c)\right]_n = e^{ax}(a^2+b^2)^{n/2}\cos\left[bx + c + n\tan^{-1}\frac{b}{a}\right]$$

So $$(e^x \cos x)_n = e^x\, 2^{n/2} \cos\left(x + \frac{n\pi}{4}\right).$$

$$\left[e^x \cos x \cdot x^2\right]_n = (e^x\cos x)_n \cdot x^2 + n\,(e^x\cos x)_{n-1} \cdot 2x + \frac{n(n-1)}{\cancel{2}}(e^x\cos x)_{n-2} \cdot \cancel{2}.$$

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$$= 2^{n/2} e^x \cos\left(x + \frac{n\pi}{4}\right) \cdot x^2 + n \cdot 2^{\frac{n-1}{2}} e^x \cos\left[x + \frac{(n-1)\pi}{4}\right]$$
$$+ n(n-1) \cdot 2^{\frac{n-2}{2}} e^x \cos\left[x + \frac{(n-2)\pi}{4}\right]$$

---

17. If $x = \tan(\log y)$ prove:

$$(1+x^2)\, y_{n+1} + (2nx - 1)\, y_n + n(n-1)\, y_{n-1} = 0.$$

Sol. $x = \tan(\log y) \Rightarrow \tan^{-1} x = \log y \Rightarrow y = e^{\tan^{-1} x}$

Now proceed as usual.

PARTIAL DERIVATIVES

NOTATION : $z = f(x,y)$, the partial derivative f_x and f_y are denoted by

$$f_x(x,y) = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = z_x$$

$$f_y(x,y) = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = z_y$$

The value of the partial derivative of $f(x,y)$ at the point (a,b) are denoted by

$$\left.\frac{\partial f}{\partial x}\right|_{(a,b)} = f_x(a,b) \quad \& \quad \left.\frac{\partial f}{\partial y}\right|_{(a,b)} = f_y(a,b)$$

---

ex: Let $z = x^2 \sin(3x + y^3)$

(a) Evaluate $\left.\frac{\partial z}{\partial x}\right|_{(\pi/3,\, 0)}$

$$\frac{\partial z}{\partial x} = 2x \sin(3x + y^3) + x^2 \cos(3x + y^3)(3)$$

$$= 2x \sin(3x + y^3) + 3x^2 \cos(3x + y^3)$$

Thus,

$$\left.\frac{\partial z}{\partial x}\right|_{(\pi/3,\, 0)} = -\frac{\pi^2}{3}$$

---

ex **IMPLICIT DIFF.**

\# $x^2 z + y z^3 = x$

Determine $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

Soln : Diff Implicitly w.r.t x

$$2xz + x^2 \frac{\partial z}{\partial x} + 3yz^2 \frac{\partial z}{\partial x} = 1$$

w.r.t to y: $x^2 \frac{\partial z}{\partial y} + z^3 + 3yz^2 \frac{\partial z}{\partial y} = 0$ 2.

PARTIAL DERIVATIVE AS A SLOPE of a Tangent line

→ line ∥ to the xz-plane and tgt to the surface $z = f(x,y)$ at the point (x_0, y_0, z_0) has SLOPE $f_x(x_0, y_0)$

→ line ∥ to the yz-plane and tgt to the surface $z = f(x,y)$ at P_0 has slope $f_y(x_0, y_0)$

---

ex $\quad z = x\sqrt{x+y}$ at $(1,3,2)$

$z = f(x,y)$

Slope of line ∥ to xz-plane:

$$f_x(x,y) = x\left(\frac{1}{2}\right)(x+y)^{-1/2}(1+0) + (1)(x+y)^{1/2} = \frac{x}{2\sqrt{x+y}} + \sqrt{x+y}$$

at $f_x(1,3) = \frac{1}{2\sqrt{1+3}} + \sqrt{1+3} = \frac{9}{4}$

PARTIAL DERIVATIVE AS RATE OF CHANGE

As the point (x,y) moves from the fixed point $P_0(x_0, y_0)$, the function $f(x,y)$ changes at a rate by $f_x(x_0, y_0)$ in the direction of positive Axis. and $f_y(x_0, y_0)$ in the direction of +ve Y-Axis.

ex: Compute the slope of the tgt line to the graph of f at the given point P_0 in the direction ∥ to the (i) xz plane (ii) yz-plane. 3)

(i) $f(x,y) = xy^3 + x^3y$: $P_0(1,-1,-2)$

Slope ∥ to xz plane is $f_x(x_0, y_0)$

$$f_x(x,y) = y^3 + 3x^2y \Rightarrow f_x(1,-1) = -1-3 = -4$$

Slope ∥ to yz plane is $f_y(x_0, y_0)$

$$f_y(x,y) = 3xy^2 + x^3 \Rightarrow f_y(1,-1) = 3+1 = 4$$

HIGHER-ORDER PARTIAL DERIVATIVES

Given $Z = f(x,y)$

Second-order partial derivatives

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = (f_x)_x = f_{xx}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = (f_y)_y = f_{yy}$$

Mixed second-order derivatives

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = (f_y)_x = f_{yx}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = (f_x)_y = f_{xy}$$

EQUALITY OF MIXED PARTIALS

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad (f_{yx} = f_{xy})$$

and

$$\frac{\partial^2 T}{\partial x^2} = \frac{\partial}{\partial x}\left(-\frac{1}{c}e^{-t}\sin\frac{x}{c}\right)$$
$$= -\frac{1}{c^2}e^{-t}\cos\frac{x}{c}$$

Thus, T satisfies the heat equation $\frac{\partial T}{\partial t} = c^2\frac{\partial^2 T}{\partial x^2}$. ■

Analogous definitions can be made for functions of more than two variables. For example,

$$f_{zzz} = \frac{\partial^3 f}{\partial z^3} = \frac{\partial}{\partial z}\left[\frac{\partial}{\partial z}\left(\frac{\partial f}{\partial z}\right)\right] \quad \text{or} \quad f_{xyz} = \frac{\partial^3 f}{\partial z\,\partial y\,\partial x} = \frac{\partial}{\partial z}\left[\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\right]$$

EXAMPLE 10 Higher-order partial derivatives of a function of several variables

By direct calculation, show that $f_{xyz} = f_{yzx} = f_{zyx}$ for the function $f(x, y, z) = xyz + x^2y^3z^4$.

Solution

First, compute the partials:

$$f_x(x, y, z) = yz + 2xy^3z^4$$
$$f_y(x, y, z) = xz + 3x^2y^2z^4$$
$$f_z(x, y, z) = xy + 4x^2y^3z^3$$

Next, determine the mixed partials:

$$f_{xy}(x, y, z) = (yz + 2xy^3z^4)_y = z + 6xy^2z^4$$
$$f_{yz}(x, y, z) = (xz + 3x^2y^2z^4)_z = x + 12x^2y^2z^3$$
$$f_{zy}(x, y, z) = (xy + 4x^2y^3z^3)_y = x + 12x^2y^2z^3$$

Finally, obtain the required higher mixed partials:

$$f_{xyz}(x, y, z) = (z + 6xy^2z^4)_z = 1 + 24xy^2z^3$$
$$f_{yzx}(x, y, z) = (x + 12x^2y^2z^3)_x = 1 + 24xy^2z^3$$
$$f_{zyx}(x, y, z) = (x + 12x^2y^2z^3)_x = 1 + 24xy^2z^3$$ ■

11.3 PROBLEM SET

1. **WHAT DOES THIS SAY?** What is a partial derivative?
2. **Exploration Problem** Describe two fundamental interpretations of the partial derivatives $f_x(x, y)$ and $f_y(x, y)$.

Determine f_x, f_y, f_{xx}, and f_{yx} in Problems 3–8.

3. $f(x, y) = x^3 + x^2y + xy^2 + y^3$
4. $f(x, y) = (x + xy + y)^3$
5. $f(x, y) = \frac{x}{y}$
6. $f(x, y) = xe^{xy}$
7. $f(x, y) = \ln(2x + 3y)$
8. $f(x, y) = \sin x^2y$

Determine f_x and f_y in Problems 9–16.

9. a. $f(x, y) = (\sin x^2)\cos y$ b. $f(x, y) = \sin(x^2\cos y)$
10. a. $f(x, y) = (\sin\sqrt{x})\ln y^2$ b. $f(x, y) = \sin(\sqrt{x}\ln y^2)$
11. $f(x, y) = \sqrt{3x^2 + y^4}$
12. $f(x, y) = xy^2\ln(x + y)$
13. $f(x, y) = x^2e^{x+y}\cos y$
14. $f(x, y) = xy^3\tan^{-1}y$
15. $f(x, y) = \sin^{-1}(xy)$
16. $f(x, y) = \cos^{-1}(xy)$

Determine f_x, f_y, and f_z in Problems 17–22.

17. $f(x, y, z) = xy^2 + yz^3 + xyz$
18. $f(x, y, z) = xye^z$
19. $f(x, y, z) = \dfrac{x + y^2}{z}$
20. $f(x, y, z) = \dfrac{xy + yz}{xz}$
21. $f(x, y, z) = \ln(x + y^2 + z^3)$
22. $f(x, y, z) = \sin(xy + z)$

In Problems 23–28, determine $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ by differentiating implicitly.

23. $\dfrac{x^2}{9} - \dfrac{y^2}{4} + \dfrac{z^2}{2} = 1$
24. $3x^2 + 4y^2 + 2z^2 = 5$
25. $3x^2y + y^3z - z^2 = 1$
26. $x^3 - xy^2 + yz^2 - z^3 = 4$
27. $\sqrt{x} + y^2 + \sin xz = 2$
28. $\ln(xy + yz + xz) = 5$ $(x > 0, \quad y > 0, \quad z > 0)$

In Problems 29–32, compute the slope of the tangent line to the graph of f at the given point P_0 in the direction parallel to

a. the xz-plane *b. the yz-plane*

29. $f(x, y) = xy^3 + x^3y$; $P_0(1, -1, -2)$
30. $f(x, y) = \dfrac{x^2 + y^2}{xy}$; $P_0(1, -1, -2)$
31. $f(x, y) = x^2 \sin(x + y)$; $P_0\left(\frac{\pi}{2}, \frac{\pi}{2}, 0\right)$
32. $f(x, y) = x \ln(x + y^2)$; $P_0(e, 0, e)$

B 33. Determine f_x and f_y for

$$f(x, y) = \int_x^y (t^2 + 2t + 1)\, dt$$

Hint: Review the second fundamental theorem of calculus.

34. Determine f_x and f_y for

$$f(x, y) = \int_{x^2}^{2y} (e^t + 3t)\, dt$$

Hint: Review the second fundamental theorem of calculus.

A function $f(x, y)$ is said to be harmonic on the open set S if f_{xx} and f_{yy} are continuous and

$$f_{xx} + f_{yy} = 0$$

throughout S. Show that each function in Problems 35–38 is harmonic on the given set.

35. $f(x, y) = 3x^2y - y^3$; S is the entire plane.
36. $f(x, y) = \ln(x^2 + y^2)$; S is the plane with the point $(0, 0)$ removed.
37. $f(x, y) = e^x \sin y$; S is the entire plane.
38. $f(x, y) = \sin x \cosh y$; S is the entire plane.
39. For $f(x, y) = \cos xy^2$, show $f_{xy} = f_{yx}$.
40. For $f(x, y) = (\sin^2 x)(\sin y)$, show $f_{xy} = f_{yx}$.
41. Find $f_{xzy} - f_{yzx}$, where $f(x, y, z) = x^2 + y^2 - 2xy \cos z$.
42. Two commodities Q_1 and Q_2 are said to be *substitute commodities* if an increase in the demand for either results in a decrease in the demand of the other. Let $D_1(p_1, p_2)$ and $D_2(p_1, p_2)$ be the demand functions for Q_1 and Q_2, respectively, where p_1 and p_2 are the respective unit prices for the commodities.
    a. Explain why $\dfrac{\partial D_1}{\partial p_1} < 0$ and $\dfrac{\partial D_2}{\partial p_2} < 0$.
    b. Are $\dfrac{\partial D_1}{\partial p_2}$ and $\dfrac{\partial D_2}{\partial p_1}$ positive or negative? Explain.
    c. Give examples of substitute commodities.
43. Two commodities Q_1 and Q_2 are said to be *complementary commodities* if a decrease in the demand for either results in a decrease in the demand of the other. Let $D_1(p_1, p_2)$ and $D_2(p_1, p_2)$ be the demand functions for Q_1 and Q_2, respectively, where p_1 and p_2 are the respective unit prices for the commodities.
    a. Is it true that $\dfrac{\partial D_1}{\partial p_1} < 0$ and $\dfrac{\partial D_2}{\partial p_2} < 0$? Explain.
    b. Determine whether $\dfrac{\partial D_1}{\partial p_2}$ and $\dfrac{\partial D_2}{\partial p_1}$ are positive or negative? Explain.
    c. Give examples of complementary commodities.
44. **Modeling Problem** The flow (in cm^3/s) of blood from an artery into a small capillary can be modeled by

$$F(x, y, z) = \frac{c\pi x^2}{4}\sqrt{y - z}$$

for constant $c > 0$, where x is the diameter of the capillary, y is the pressure in the artery, and z is the pressure in the capillary. Compute the rate of change of the flow of blood with respect to
    a. the diameter of the capillary
    b. the arterial pressure
    c. the capillary pressure
45. **Modeling Problem** Biologists have studied the oxygen consumption of certain furry mammals. They have found that if the mammal's body temperature is T degrees Celsius, fur temperature is t degrees Celsius, and the mammal does not sweat, then its relative oxygen consumption can be modeled by

$$C(m, t, T) = \sigma(T - t)m^{-0.67}$$

(kg/h), where m is the mammal's mass (in kg) and σ is a physical constant. Compute the rate (rounded to two decimal places) at which the oxygen consumption changes with respect to
    a. the mass m
    b. the body temperature T
    c. the fur temperature t
46. **Modeling Problem** A gas that gathers on a surface in a condensed layer is said to be *adsorbed* on the surface, and the surface is called an *adsorbing surface*. The amount of gas adsorbed per unit area on an adsorbing surface can be modeled by

$$S(p, T, h) = ape^{h/(bT)}$$

where p is the gas pressure, T is the temperature of the gas, h is the heat of the adsorbed layer of gas, and a and b are physical constants. Compute the rate of change of S with respect to
    a. p b. h c. T
47. The *ideal gas law* says that $PV = kT$, where P is the pressure of a confined gas, V is the volume, T is the temperature, and k is a physical constant.